

Max vs Min Spanning Tree

Spanning Trees, Greedy on Sorted Edges · Mentor-Led Lab (Student Edition)

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Difficulty ★★★ Advanced

Problem

Compute both the minimum and the maximum spanning tree weight of a connected weighted graph, and their difference. The maximum spanning tree is Kruskal's algorithm run on edges sorted from heaviest to lightest.

Implement both with the same union-find core.

What to Compute

- Max Spanning Weight: the total weight of the maximum spanning tree.
- Min Spanning Weight: the total weight of the minimum spanning tree.
- Difference: max minus min.

Input / Output Format

Input: Line 1: N M . Next M lines: an undirected weighted edge u v w .

Output: Three lines: Max Spanning Weight, Min Spanning Weight, Difference.

Examples

Example 1

Input: 4 5 0 1 1 0 2 3 1 2 1 1 3 4 2 3 2

Output:

Max Spanning Weight: 9
Min Spanning Weight: 4
Difference: 5

Explanation: The minimum spanning tree weighs 4; choosing heaviest-first (4, 3, 2) gives a maximum spanning tree of 9, a difference of 5.

Example 2

Input: 3 3 0 1 1 1 2 2 0 2 3

Output:

Max Spanning Weight: 5
Min Spanning Weight: 3

Difference: 2

Explanation: Min tree $1+2 = 3$; max tree $3+2 = 5$; difference 2.

Constraints

- Sort edges by (weight, u, v); reverse the order for the maximum tree.
- Add an edge only if it joins two different groups.
- Assume the graph is connected.

Sample Run

Sample Input

```
4 5
0 1 1
0 2 3
1 2 1
1 3 4
2 3 2
```

Sample Output

```
Max Spanning Weight: 9
Min Spanning Weight: 4
Difference: 5
```